HDOC- 7 – 1
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Program to Check Whether a Year is a Leap Year Using Conditional Statements

This document provides a clean, well-commented implementation for checking whether a given year is a leap year in C. The program takes a year as input and uses an if–else if–else ladder with the modulus operator to decide and display the result.

1. C Implementation

Step 1: Understand the Logic

- A year divisible by 400 is a leap year (for example 2000).
- A year divisible by 100 but not by 400 is not a leap year (for example 1900).
- A year divisible by 4 but not by 100 is a leap year (for example 2024).
- Every other year is not a leap year (for example 2023).

Step 2: Define Variables and Data Types

- Use an integer variable (int) to store the year.

Step 3: Take User Input

- Display a prompt using printf("Enter a year: "), then read the value using scanf("%d", &year).

Step 4: Apply the Conditional Statements

1. If year % 400 == 0, it is a leap year.
2. Else if year % 100 == 0, it is not a leap year.
3. Else if year % 4 == 0, it is a leap year.
4. Otherwise (else), it is not a leap year.

Step 5: Format the Output

- Print the year along with the message "a leap year" or "not a leap year" using printf().

Complete C Code

```c
/*
 * Q8 (User Inputs, Operations & Output)
 * Write a program to input a year and check whether it is
 * a leap year or not using conditional statements.
 */
#include <stdio.h>

int main() {
    int year;

    // Step 1: Read the year
    printf("Enter a year: ");
    scanf("%d", &year);

    // Step 2: Check the leap year conditions
    if (year % 400 == 0) {
        printf("%d is a leap year.\n", year);
    } else if (year % 100 == 0) {
        printf("%d is not a leap year.\n", year);
    } else if (year % 4 == 0) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc7_1.c -o hdoc7_1
$ ./hdoc7_1
Enter a year: 2024
2024 is a leap year.
```

Run 2:

```
$ ./hdoc7_1
Enter a year: 1900
1900 is not a leap year.
```

Run 3:

```
$ ./hdoc7_1
Enter a year: 2000
2000 is a leap year.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: 2024
- Calculation:
  - 2024 % 400 = 24, not 0, so the first condition is FALSE
  - 2024 % 100 = 24, not 0, so the second condition is FALSE
  - 2024 % 4 = 0, so the third condition is TRUE
- Output: 2024 is a leap year.

Sample Test Case 2:

- Input: 1900
- Calculation:
  - 1900 % 400 = 300, not 0, so the first condition is FALSE
  - 1900 % 100 = 0, so the second condition is TRUE
- Output: 1900 is not a leap year.

Sample Test Case 3:

- Input: 2000
- Calculation:
  - 2000 % 400 = 0, so the first condition is TRUE
- Output: 2000 is a leap year.